



Bellabeat : Smart Device Usage & Wellness Behaviour Analysis

Analyzing Fitbit user behaviour to identify opportunities for Bellabeat's wellness products

DATA ANALYTICS CASE STUDY:

Tools: Excel / MySQL / Power BI

Dataset: Fitbit Fitness Tracker Data

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Executive Summary

This project analyzes Fitbit fitness-tracker data to understand user behavior across daily activity, sleep, heart rate, and weight tracking. The analysis was performed using MySQL.

The analysis identified several important behavioral patterns:

- **Light activity dominates:** 84.89% of recorded active minutes were light activity, while only 8.97% were very active.
- **Prolonged inactivity is common:** 3,195 inactive sessions lasted 90+ minutes, creating an opportunity for personalized movement reminders.
- **Sleep tracking has meaningful adoption:** Sleep data was available for 25 users, while activity data covered 35 users, showing that sleep is a significant part of the tracked wellness behavior.
- **Weight tracking has lower adoption:** Only 13 users had weight-log records, with 10 of those users manually recording their weight, suggesting an opportunity to make weight tracking easier and more engaging.
- **Short sleep affects a significant subset:** Among users with more than 15 qualifying night-sleep sessions, 47.06% showed moderate or frequent short-sleep patterns.
- **Short sleep is associated with less consistent sleep timing:** Frequent short sleepers showed greater variation in both bedtime and wake-up time than infrequent short sleepers.
- **Weekends show a shift in both activity and sleep:** Average weekend steps were 7,188 compared with 7,453 on weekdays, while users slept about 18 minutes longer, went to bed about 13 minutes later, and woke up about 31 minutes later on weekends.
- **Heart rate increases with activity intensity:** Average heart rate increased from 68.48 BPM during sedentary activity to 120.17 BPM during very active activity. However, heart-rate data was available for only 15 of 35 users.

Business Task

Use Fitbit behavioral data to identify recurring user patterns and translate them into actionable Bellabeat product opportunities, without making medical or causal claims unsupported by the dataset.

- How active are users and what type of activity dominates?
- Where do prolonged inactivity and movement opportunities occur?
- How does activity intensity relate to heart rate?
- How complete and useful is the available heart-rate data?
- How long and how consistently do users sleep?
- Do recurring short-sleep patterns overlap with irregular sleep schedules?

- How does sleep behavior differ between weekdays and weekends?
- What evidence exists for sleep efficiency and sleep fragmentation?
- How widely are weight-tracking features used?
- Which findings can become practical Bellabeat recommendations?

Data and Analytical Scope

The analysis covered daily activity, hourly/minute activity, sleep, heart rate, and weight-log data from the Fitbit Fitness Tracker dataset used in the Bellabeat case-study context.

Data area	Purpose
Daily activity	Steps, distance, activity minutes, sedentary behavior, calories
Hourly / minute activity	Time-of-day behavior and detailed activity patterns
Sleep	Duration, onset/wake consistency, short-sleep patterns, efficiency and awake/restless states
Heart rate	HR profile and relationships with activity intensity and sleep
Weight log	Tracking coverage and manual-report adoption

Data Cleaning and Preparation

The main EDA SQL file focuses only on exploratory analysis and business insights. The complete data-loading, cleaning, validation, and integration process is not included in this.

1. Data Loading:

- Imported the Fitbit CSV files into MySQL raw/staging tables using (LOAD DATA LOCAL INFILE).
- Loaded date and time fields as strings where necessary to avoid import-format issues.
- Created separate raw tables for the different datasets before performing transformations.

2. Date and Time Cleaning

- Converted string-based dates and timestamps into proper DATE and DATETIME formats using (STR TO DATE()).
- Created cleaned date/time columns while preserving the original raw values.
- Checked the minimum and maximum dates of each dataset to understand their available time periods.
- Compared the date ranges of related datasets and identified their overlapping periods before combining them.

3. Data Validation

After loading and converting the data, the following checks were performed:

- Total number of records.
- Number of unique users.
- NULL/missing values.
- Minimum and maximum dates.
- Duplicate user-date or user-session records.
- Data ranges and unusual values.
- User coverage across different datasets.

User coverage differed between source datasets. After cleaning and integrating the relevant activity data, 35 unique users were available for the main activity analysis.

4. Data Cleaning and Integration

- Cleaned the imported data before creating the final analysis-ready tables.
- Combined relevant datasets using **Id** and, where appropriate, **date/time fields**.
- Used overlapping date periods to ensure that comparisons between related datasets were based on compatible time ranges.
- Created indexes on frequently used columns such as: (Id, ActivityDate) , (Id, ActivityHour) , (Id, ActivityMinute)

5. Sleep Data Preparation

For the minute-level sleep data:

- Used MIN(ActivityMinute) to identify the beginning of each sleep session.
- Used MAX(ActivityMinute) to identify the end of each sleep session.
- Used TIMESTAMPDIFF() to calculate sleep duration in minutes.
- Classified sessions into:
 - Afternoon sleep
 - Evening sleep
 - Normal night sleep
 - Late-night sleep
 - Other
- For the deeper night-sleep analysis, retained sessions between **4 and 12 hours** to focus on plausible night-sleep periods.
- Created additional measures such as **sleep onset time, wake-up time, and sleep duration** for consistency analysis.

6. Final Analysis Dataset

After loading, cleaning, validating, and integrating the data, the resulting tables were used for the **EDA and business analysis**.

The analysis then focused on:



Exploratory Data Analysis

1. Daily Activity Analysis

Objective: Understand users' overall activity levels and identify opportunities to encourage healthier movement patterns.

Analysis:

- Average daily steps and distance were analyzed.
- Activity was categorized into sedentary, light, moderate, and very active levels.
- Prolonged inactive periods were identified.

Key Findings:

- Light activity dominates: 84.89% of recorded active minutes were spent in light activity, while only 8.97% were very active.
- Prolonged inactivity is common: 3,195 inactive sessions lasted 90+ minutes, indicating frequent extended periods of inactivity.
- These patterns suggest an opportunity for Bellabeat to encourage more movement throughout the day, rather than focusing only on intense workouts.

2. Weekday vs Weekend Activity

Objective: Determine whether users' activity levels change between weekdays and weekends.

Analysis:

Metric	Weekday	Weekend
Average Steps	7,452.94	7,188.27
Average Distance	5.34	5.16

Key Finding:

- Weekend activity was slightly lower than weekday activity: average daily steps decreased from 7,452.94 on weekdays to 7,188.27 on weekends, a difference of approximately 265 steps per day.
- Average distance also decreased slightly from 5.34 to 5.16.
- This suggests a small reduction in overall activity during weekends.

3. Sleep Adoption and Duration

Objective: Understand how widely users track sleep and identify common sleep-duration patterns.

Analysis:

- Compared sleep-data users with the overall activity-user population.
- Analyzed the distribution of night-sleep duration.
- Identified users with recurring short sleep.

Key Findings:

- Sleep tracking had meaningful adoption: 25 of the 35 users had sleep data.
- Short sleep was relatively common: 17.09% of qualifying night-sleep sessions lasted less than 6 hours.

- Recurring short sleep affected a substantial subgroup: Among users with more than 15 qualifying night-sleep sessions, 47.06% showed moderate or frequent short-sleep patterns.
- This indicates an opportunity for Bellabeat to provide personalized sleep-duration and sleep-routine guidance rather than simply reporting total sleep time.

4.Short Sleep and Sleep Consistency

Objective: Determine whether users who frequently experience short sleep also have less consistent sleep schedules.

Analysis:

Sleep Group	Avg Short-Sleep %	Bedtime Variation	Wake-Time Variation
Frequent	48.55%	77.24 min	97.48 min
Moderate	20.79%	77.36 min	79.10 min
Infrequent	6.99%	58.02 min	72.90 min

Key Finding:

- **Frequent short sleepers had less consistent sleep schedules**, with greater variation in both bedtime and wake-up time.
- Their average **bedtime variation was 77.24 minutes**, compared with **58.02 minutes** among infrequent short sleepers.
- Their average **wake-up variation was 97.48 minutes**, compared with **72.90 minutes** among infrequent short sleepers.
- This suggests that **recurring short sleep is accompanied by greater irregularity in sleep timing** in this dataset.

5.Weekday vs Weekend Sleep

Objective: Understand how sleep behavior changes between weekdays and weekends.

Analysis

Metric	Weekday	Weekend
Average Sleep	443.89 min	462.31 min
Average Bedtime	~11:34 PM	~11:47 PM
Average Wake-up	~6:58 AM	~7:29 AM

Key Findings:

- Weekend sleep duration increased by 18.42 minutes compared with weekdays.

- Average bedtime shifted 12.79 minutes later on weekends.
- Average wake-up time shifted 31.21 minutes later, showing that users primarily extended weekend sleep by waking up later.
- Overall, users showed a clear shift toward a later and slightly longer sleep schedule on weekends, indicating a difference between weekday and weekend sleep routines.

6.Heart Rate and Activity Intensity

Objective: Understand how heart rate changes with activity intensity and evaluate the availability of HR data.

Analysis

Activity Intensity	Average Heart Rate
Sedentary	68.48 BPM
Light	85.98 BPM
Moderate	101.91 BPM
Very Active	120.17 BPM

Key Findings:

- Heart rate increased steadily with activity intensity, rising from 68.48 BPM during sedentary activity to 85.98 BPM during light activity, 101.91 BPM during moderate activity, and 120.17 BPM during very active activity.
- This shows a clear relationship between activity intensity and heart-rate response, making heart rate useful for personalized activity feedback.
- Heart-rate data was available for only 15 of the 35 users, meaning HR-based insights apply to a smaller subset of the users.
- Therefore, heart-rate features should complement the core activity experience rather than be required for all users.

7.Weight Tracking Adoption

Objective: Understand how many users actively track their weight and how they record it.

Analysis:

- 13 of 35 users had weight-log records.
- 10 of the 13 users recorded their weight manually.

Key Findings:

Weight tracking adoption was relatively low compared with activity tracking. Only 13 of 35 users had weight-log records, and most of those users manually recorded their weight. This suggests that weight tracking was used by a smaller subset of users and may represent an opportunity to improve the convenience and engagement of weight-related features

Recommended Bellabeat Product

Based on the analysis of Fitbit user activity, sleep, heart rate, and weight-tracking data, the Bellabeat Leaf was identified as the most suitable product for recommendation.

The analysis revealed that users frequently engage in light activity, experience prolonged periods of inactivity, and exhibit inconsistent sleep behaviors. Since **Bellabeat Leaf** focuses on activity tracking, sleep monitoring, and wellness insights, it is well positioned to address these user behaviors.

Actionable Recommendations for Bellabeat Leaf

1. Personalized Movement Reminders

Finding:

- 84.89% of active minutes were light activity.
- Only 8.97% were very active.
- 3,195 inactive sessions lasted 90+ minutes.

Recommendation:

Introduce personalized movement reminders that detect prolonged inactivity and adapt to the user's daily routine instead of using fixed hourly alarms.

Marketing Strategy:

"Your Friendly Wellness Companion"

Use empathetic, contextual, and gamified notifications that turn long inactive periods into small achievable challenges.

Example:

"You've been focused for 2 hours. Take a 2-minute movement break!"

Business Value:

- Increase user engagement
- Encourage regular movement
- Reduce prolonged inactivity
- Strengthen Bellabeat's personalized wellness experience

2. Progressive Activity Challenges

Finding:

- 84.89% of active minutes were light activity.
- Only 8.97% were very active.

Recommendation:

Introduce personalized activity challenges that gradually increase activity goals based on each user's current activity level and progress.

Marketing Strategy:

"Smart Progression, Not Overwhelming Targets"

Use adaptive goals, milestone badges, and small challenges to make users feel that their progress is achievable and rewarding.

Example:

"You've hit your goal 4 days in a row! Ready for your next challenge?"

Business Value:

- Encourage users to gradually increase activity.
- Improve motivation through achievable goals and rewards.
- Increase long-term engagement and retention.

3. Sleep Consistency Coaching

Finding:

Frequent short sleepers showed greater variation in bedtime and wake-up times than infrequent short sleepers.

Recommendation:

Provide personalized sleep-consistency reminders and bedtime guidance based on each user's sleep patterns.

Marketing Strategy:

"Sleep Better Through Consistency"

Use gentle, personalized wind-down reminders and encourage users to maintain a more consistent sleep schedule rather than focusing only on total sleep hours.

Example:

"Your body loves a routine. Time to start winding down for tonight "

Business Value:

- Encourage more consistent sleep routines.
- Help users understand their sleep patterns.
- Increase engagement with Bellabeat's sleep-tracking features.

4. Weekly Wellness Summary

Finding:

Users showed clear differences between weekday and weekend activity and sleep patterns.

Recommendation:

Provide a simple weekly wellness summary combining movement and sleep insights in one easy-to-understand report.

The summary could include:

- Average daily steps
- Prolonged inactivity periods
- Average sleep duration
- Short-sleep nights
- Sleep consistency
- Weekday vs. weekend comparison

Marketing Strategy:

"Understand Your Week at a Glance"

Use personalized weekly insights to help users recognize their habits, celebrate progress, and identify areas for improvement.

Example:

"Your week at a glance: You moved more on weekdays, but caught up on sleep this weekend."

Business Value

- Helps users understand their overall wellness patterns.
- Encourages continued engagement with the Bellabeat app.
- Connects **activity and sleep insights** in one place.
- Creates a recurring reason for users to return to the app.

Conclusion

The analysis of Fitbit fitness-tracker data provided valuable insights into users' activity, sleep, heart rate, and wellness-tracking behavior.

The findings show that users generally engage in light activity, experience periods of prolonged inactivity, and show noticeable differences in weekday and weekend routines. Sleep analysis also revealed a meaningful group of users with recurring short sleep and less consistent sleep schedules. Heart-rate analysis showed a clear relationship between activity intensity and heart rate, while the relatively low adoption of weight tracking highlighted another opportunity for improvement.

Based on these patterns, Bellabeat can move beyond simply displaying health data and focus on turning tracked behavior into personalized guidance. The recommended features—Personalized Movement Reminders, Progressive Activity Challenges, Sleep Consistency Coaching, and Weekly Wellness Summaries—can help users better understand their habits and take small, achievable actions.

Limitations

- The analysis is based on 35 users with varying levels of tracking activity, so the findings act as a starting direction rather than a final rule for all users.
- Differences in device tracking dates and habits were cleaned and matched so overlapping trends could be compared fairly.
- The trends show clear connections between movement, sleep, and heart rate, but they reflect habits rather than direct cause and effect.
- These insights offer strong, data-backed ideas for Bellabeat that can be tested in the real app to confirm their impact.

Future Scope

- Larger user dataset: Analyze a larger and more representative Bellabeat user population.
- Personalized recommendations: Use individual user history to dynamically adjust movement and sleep recommendations.
- Real-time personalization: Combine activity, sleep, and heart-rate data to provide context-aware recommendations.
- Feature effectiveness testing: Use A/B testing to measure whether movement reminders and sleep coaching actually improve engagement and behavior.
- Predictive analytics: Develop models to identify users who are likely to experience prolonged inactivity or recurring short sleep.
- Dashboard integration: Present personalized insights and weekly wellness summaries through an interactive dashboard.